# NSSA-220 Project 1

## Application Performance Monitoring



# Application Performance Monitoring (APM)

- Monitoring application performance is done for many reasons including, but not limited to:
  - Detecting systemic problems with software, such as memory leaks
  - Determining if computing resources are sufficient for the mix of software being used (number of CPUs, network bandwidth, memory, disk space, disk speed)
  - Detecting malicious activity in a system

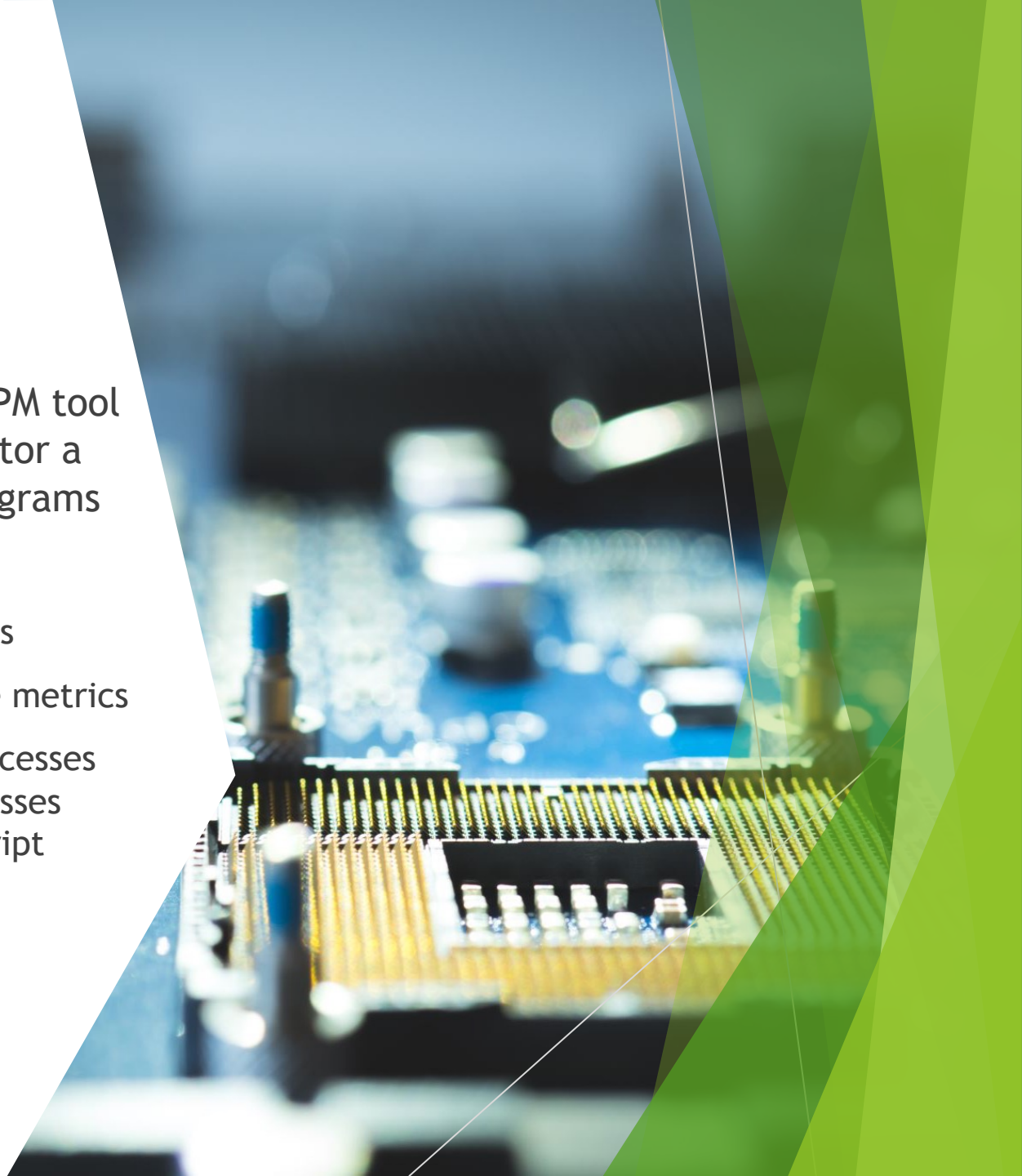
# Points of Inspiration

APM is highly relevant to just about any career path in computing, especially systems engineering, network management, network and system security, and system administration

Employers are impressed by anyone that has experience with APM and analyzing APM data is highly related to the field of data analytics/data science

# APM Tool Overview

- You'll be writing an APM tool in Bash that will monitor a mix of provided C programs
- The script will:
  - start these processes
  - collect performance metrics
  - terminate these processes and any other processes spawned by your script



# APM Tool Overview (continued)



The APM tool will monitor application performance by collecting metrics at both:

- Process level
- System level



Your final submission will include a short report with Excel plots of the performance metrics collected and a short analysis and conclusions

# Process Level Metrics



Linux provides the ability to collect performance metrics at the process level for both CPU and memory



If the applications are utilizing most of or all of the CPU, you may consider adding cores to a VM or processors to a physical machine

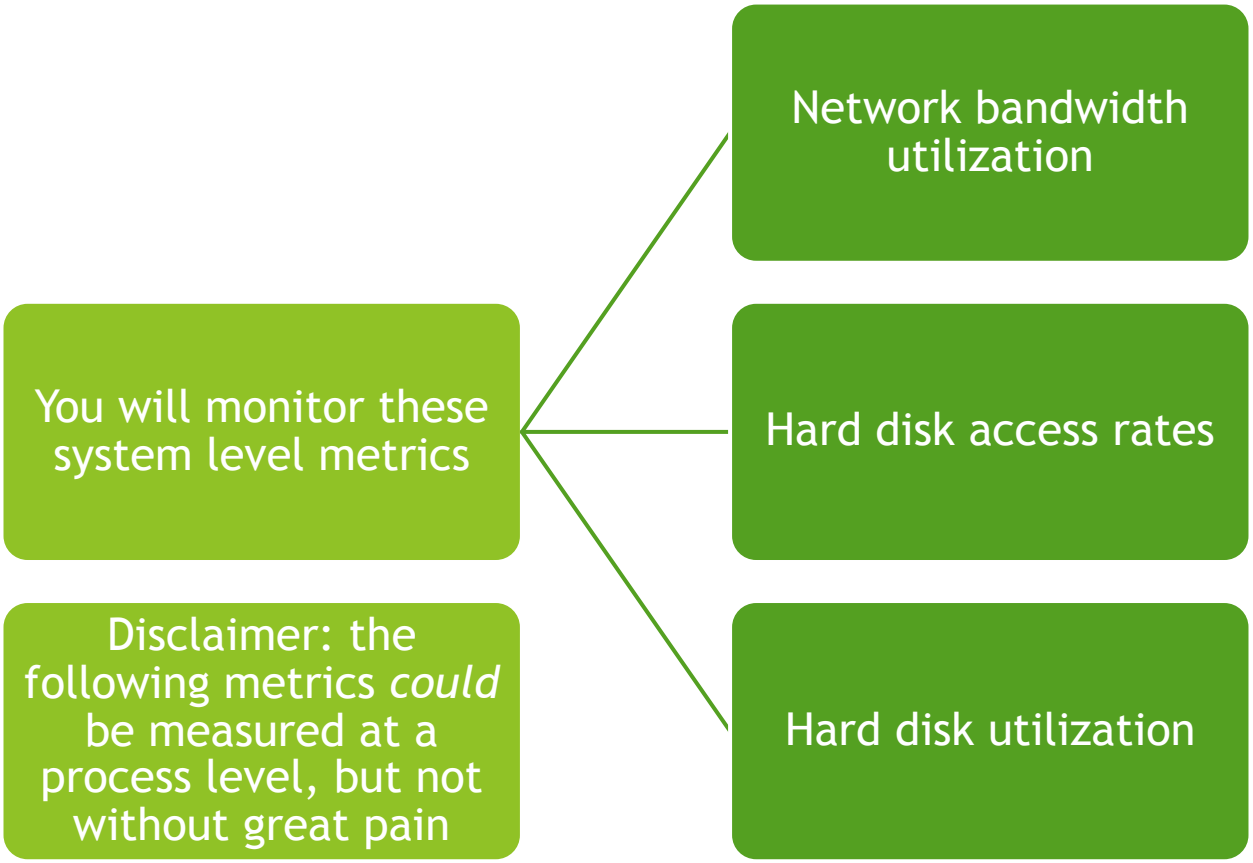


Unbounded memory growth over time indicates poor memory management



Using *ps*, you can see the percent of CPU and Memory that a process is using

# System Level Metrics



# Network Bandwidth Utilization



Network bandwidth utilization indicates how much bandwidth each network interface card is using over time



If data rates are approaching network capacity, additional capacity and/or load balancing is likely needed



You will measure bandwidth utilization in terms of receive (RX) data rate and transmit (TX) data rate using the *ifstat* tool



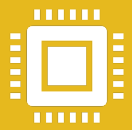
# Hard Disk Access Rates

- Hard disk access rates are typically measured in volume of reads and writes per second
- Monitoring disk access activity over time may show that a solid-state drive is needed over a mechanical drive or alert a system administrator to unexpected excessive access rates
- You will measure hard disk writes in kB/second to the primary hard drive (sda) using the *iostat* tool

# Hard Disk Utilization



Even with the current enormous amounts of disk capacity, disk utilization is still a challenging problem



You will measure hard disk utilization on the “/” mount (the /dev/mapper/rl-root filesystem) using the *df* tool

# Applications to Monitor



You will be given 6 applications in the form of pre-compiled C executables



Each application requires a single argument: the "real" IP address (not `127.0.0.1`) assigned to the NIC, determined from the *ifconfig* command



On VMWare, verify that the VM network is in “NAT Mode” and is enabled.

# Utilities for Metrics Collection

| Utility | Purpose |
| --- | --- |
| ps | Collects % CPU and % Memory utilization |
| ifstat | Measures network bandwidth utilization |
| iostat | Measures hard disk access rates |
| df | Monitors hard disk utilization |

# Project 1 Information

| Done in teams of 3 | | |
|---|---|---|
| Declare your team on myCourses discussion area | Yes, you must do this in teams! No lone wolves allowed! | Teamwork and participation will impact your grade! |

Projects are designed to be more difficult than labs and require your own research beyond what you've been taught so far (worth 15% of your grade)

# APM Tool Metrics Requirements

1. The APM tool shall collect process and system level metrics every 5 seconds
2. The APM tool shall collect % CPU and % memory utilization *per process* **using the ps tool**
3. The APM tool shall collect network bandwidth utilization in terms of RX data rate and TX data rate (kB/s) with a *sampling interval of 1 second* on the ens interface **using the ifstat tool**
4. The APM tool shall collect hard disk writes in kB/second to the primary hard drive (sda) **using the iostat tool**
5. The APM tool shall collect hard disk utilization of the "/" mount in *Megabytes available* **using the df tool**

# APM Tool Output Requirements

6. The APM tool shall output all **process level** CPU and memory metrics to a CSV file specific to the process they were measured from called `<proc_name>_metrics.csv`
7. The format of the process specific output files shall be:
   - <seconds since script start>, <%CPU>, <%memory>
8. The APM tool shall write all **system level** metrics to a single file called `system_metrics.csv`
9. The format of the system level output file shall be:
   - <seconds since script start>, <RX data rate>, <TX data rate>, <disk writes>, <available disk capacity>

# APM Tool Other Requirements

10. The APM tool shall minimally include 3 functions to (1) spawn applications and other processes, (2) collect process level metrics, and (3) collect system level metrics
11. The APM tool shall spawn all application and other necessary processes
12. The IP address can be hardcoded in the script (but will need to be edited if run on a different system)
13. The APM tool shall kill all application processes and any other processes it spawns in an additional exit trap function called "`cleanup`"
14. You will run the tool to gather metrics for **15 minutes** and manually terminate the script. You should not write code to terminate the script after 15 minutes
15. Your team's submission to MyCourses shall be a *single zip file* including: your APM tool script, all output files from your 15-minute run, and a report with Excel plots of the metrics collected (matching the provided format)

# Project Submission

You will submit a single Zip file to MyCourse for the team that includes:

- Your APM tool script
- Six process metrics and one system metrics output files from your 15-minute run
- Your completed report using the provided format with Excel plots of the metrics collected and your analysis as indicated

# Grading

- See grading sheet for details
- Major point loss will occur if an individual exhibits poor teamwork. You will fill out a peer review form to indicate how well the team worked together.
- Let me know ASAP if there are issues with your team.



# Linux VM Setup

## Linux VM resources:

| 1 CPU core | 4 GB memory | Default hard disk size | Rocky NIC in NAT mode (ens???) |
|---|---|---|---|

Install iostat: `sudo dnf install -y sysstat`

When generating your data you should use above resource limits.
Your development can use any Rocky Linux configuration

# Hints



Read the man pages for the specified tools. Don't rely on Google as options vary for different Operating Systems!



One of the tools requires you to spawn an additional process and monitor its output to obtain metrics properly.



All the values in the output files should be numbers only as you are given what the units are.



Put all the requirements into a table in a Word document and check them off as you complete them.



There are clear characteristics of solutions that rely on LLMs. You need to understand what is in your script and the coding decisions made.

# Ask for help!

1

Don’t suffer in silence. Ask for help sooner rather than later!

- Attend office hours
- Make an appointment outside of office hours
- Send an email

2

If you’re not sure if your team has met a specific requirement, please ask!

3

Waiting until the last weekend to start the assignment will result in hardship and heartache!